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Experiment:- Study of LAN Topology, Network devices and Address Configuration.

Objective:- →

- 1) To study the LAN topology used in college campus.
- 2) To identify networking device & understand their function.
- 3) To analyze wired & wireless communication method used in the network.
- 4) To measure speed of Network for wired & wireless connections.
- 5) To obtain MAC, IP Address & subnet mask.

Problem Statement -1

• Draw & identify LAN Topology of college campus & clearly label all components such as router, switches & etc.

→ Solⁿ → The college campus network observed this study consists of:

- * Layer-3 Core switches in the server Room.
- * Managed by PoE switches.
- * Wireless Access point (APs)
- * Ethernet port in classroom & laboratories.

Network Topology

Hybrid Hierarchical Topology which combines.

- 1) Tree Topology
- 2) Extended Star Topology.
- 3) Ring Topology (Server room).

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The network hierarchical layer

- 1) Core layer.
- 2) Distribution layer.
- 3) Access layer

Core Layer

Core layer contains Server Room and server room have Firewall, Router, 2-L3+ switches, Patch Rack, Server, inverter, etc.

Distribution Layer

* Distribution layer contains 4-switches.

* The 4 switches are →

- ① Main Build → R7 (Room no-213, 2nd floor)
- ② Main Build → R1 (Room no-301, 3rd floor).
- ③ Main Build → R6 (Room no-113, 1st floor)
- ④ Main Build → R2 (Room no-201, 2nd floor).

* This 4-switches contains →

- ① 18 Port Gigabit Easy smart switch with 16-port.

TL-SG1218MPE

- ② SG3428XMP → 24 Port Gigabit & 4-port 10 GbESFP+, L2+ managed switches with 24 Port PoE+.

→ Now let's discuss these 2 above components →

- ① TP-Link TL-SG1218MPE

* Likely used for: CCTV camera, wifi access point, etc.

* 18 Port Gigabit "Easy Smart" Switch.

* 16 Port Support PoE+

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② TP-Link Omada S613428xMP

* Higher-end managed Layer 2+switch.

* 24 PoE+ port, uplinks, 10G SFP

* Usually act as a main distribution for each floor.

→ Access Layer

* Access Layer contain 3-switches i.e

1) Main Build R1 - [Room no-301, 3rd floor],

2) Main Build R4 [Room no-05, Ground floor],

3) Main Build R5 [Room no-18, Ground floor].

* Every switches act as access layer which helps to provide network connectivity to all classroom, Lab, Faculty Room, etc.

* Access Layer directly Connect:

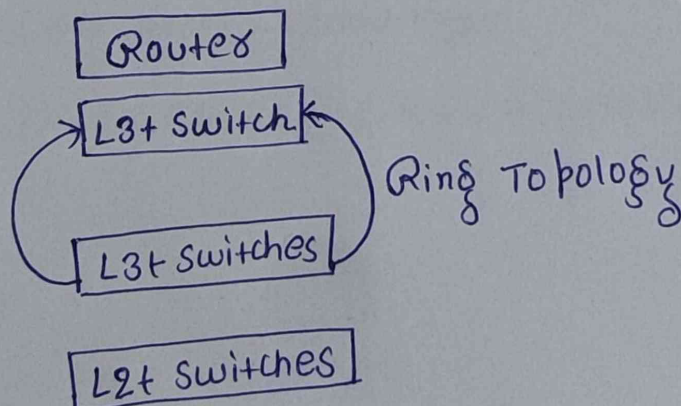
① Ethernet Port

② Wireless AP

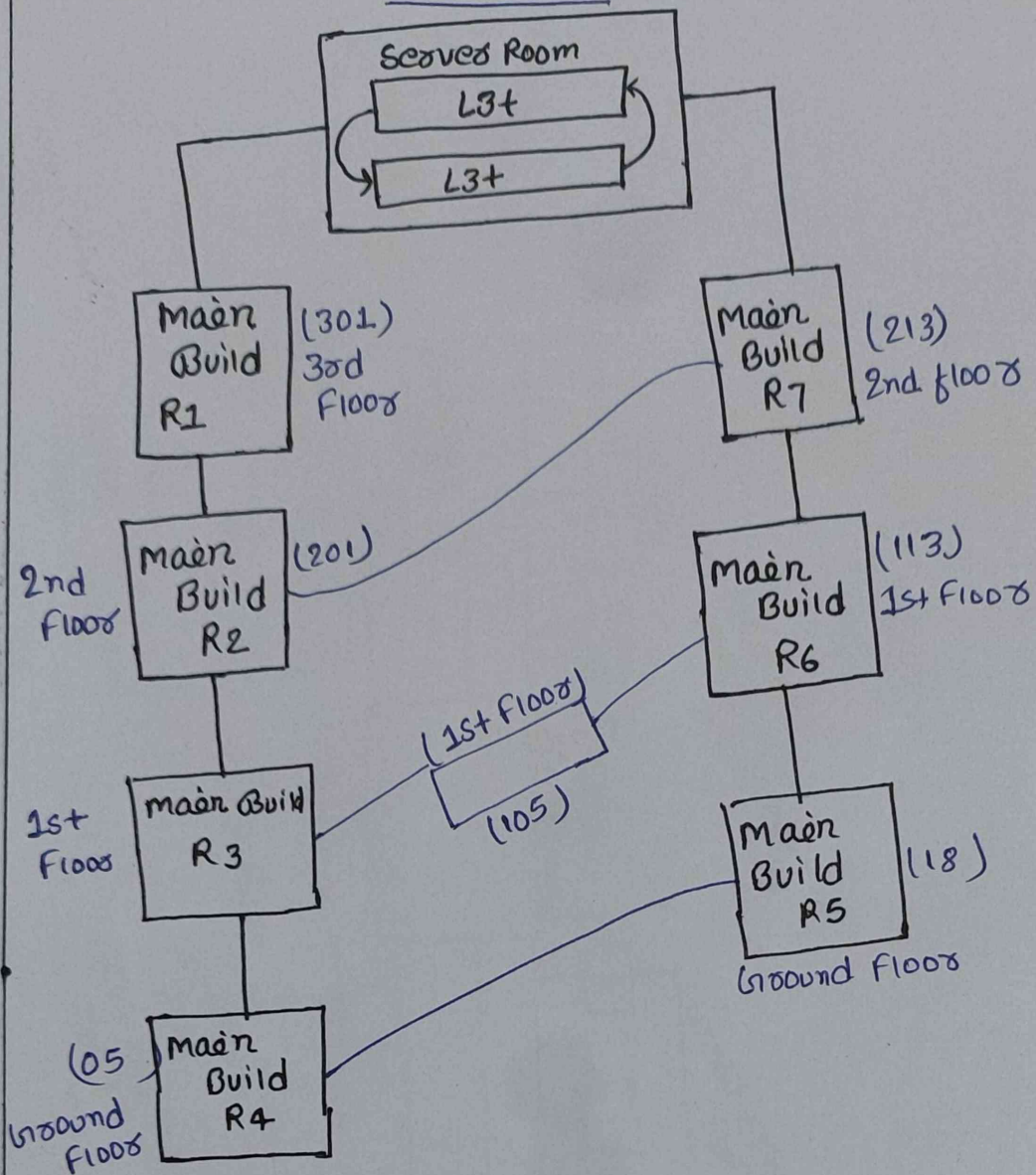
③ Lab system

④ Outdoor wireless AP (ex-D18)

Server Room Architecture / Topology.



TOPOLOGY



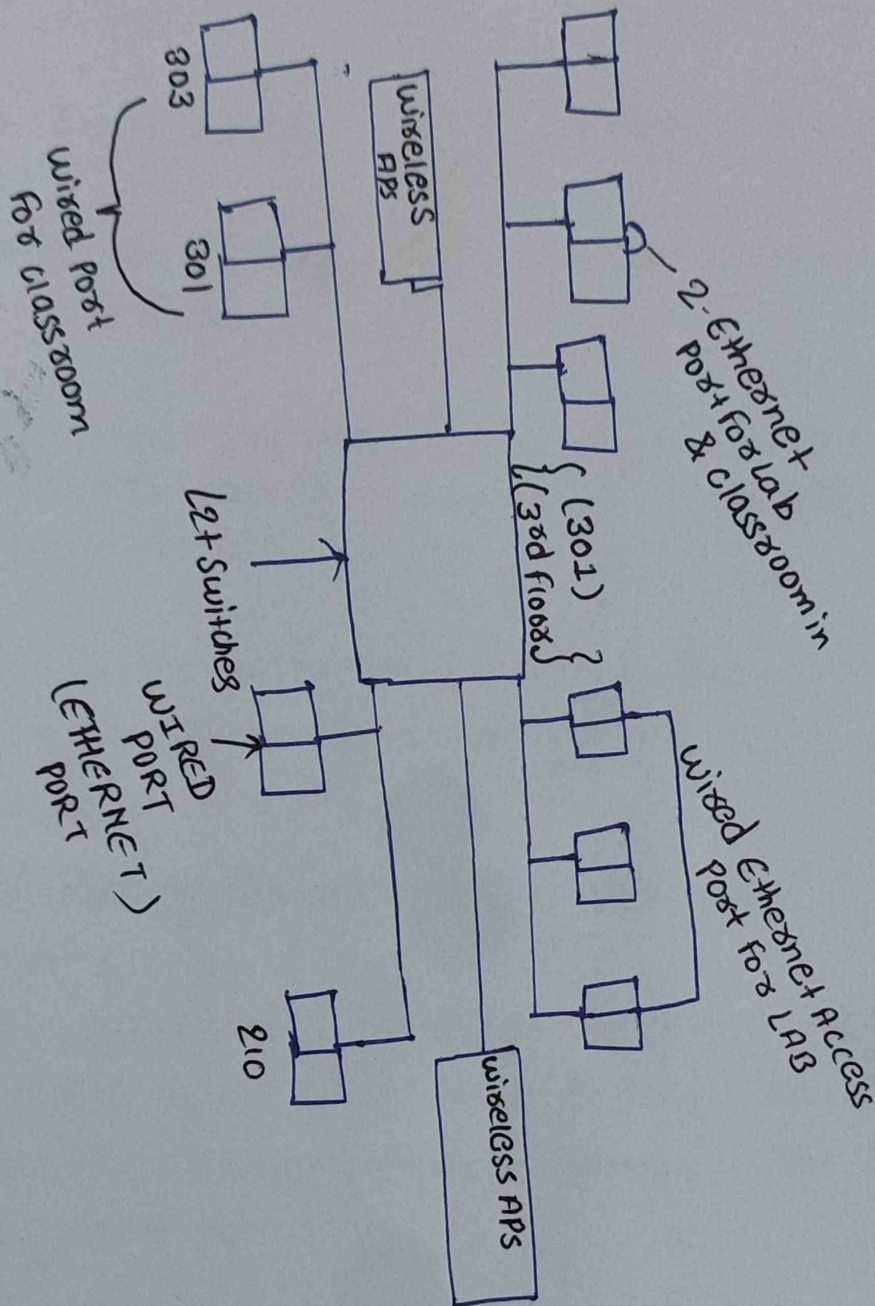
→ R1, R7, R2, R6 → Distribution layer

→ Server room → Core layer

→ R1, R2, R3, R4, R5, R6, R7 → Access layer

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ACCESS LAYER SWITCH (3rd Floor - 301)



Windows PowerShell

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Install the latest PowerShell for new features and improvements! <https://aka.ms/PSWindows>

PS C:\Users\LENOVO> netsh wlan

The following commands are available:

Commands in this context:

?	- Displays a list of commands.
add	- Adds a configuration entry to a table.
connect	- Connects to a wireless network.
delete	- Deletes a configuration entry from a table.
disconnect	- Disconnects from a wireless network.
dump	- Displays a configuration script.
export	- Saves WLAN profiles to XML files.
help	- Displays a list of commands.
IHV	- Commands for IHV logging.
refresh	- Refresh hosted network settings.
reportissues	- Generate WLAN smart trace report.
set	- Sets configuration information.
show	- Displays information.
start	- Start hosted network.
stop	- Stop hosted network.

To view help for a command, type the command, followed by a space, and then type ?.

PS C:\Users\LENOVO> netsh wlan show interface

There is 1 interface on the system:

```
PS C:\Users\LENOVO> netsh wlan show interface
```

```
There is 1 interface on the system:
```

```
Name                : Wi-Fi
Description          : Intel(R) Wi-Fi 6 AX203
GUID                 : b0dd9445-2ec8-4691-a2f9-f04a0b8ecad9
Physical address     : ae:39:d7:98:56:7d
Interface type       : Primary
State                : connected
SSID                 : GECK-Campus-WiFi
AP BSSID             : b0:19:21:74:07:b9
Band                 : 5 GHz
Channel              : 48
Connected Akm-cipher : [ akm = 00-0f-ac:02, cipher = 00-0f-ac:04 ]
Network type         : Infrastructure
Radio type           : 802.11ac
Authentication        : WPA2-Personal
Cipher                : CCMP
Connection mode       : Auto Connect
Receive rate (Mbps)   : 390
Transmit rate (Mbps)  : 433.3
Signal                : 85%
Rssi                  : -69
Profile              : GECK-Campus-WiFi
QoS MSCS Configured   : 0
QoS Map Configured    : 0
QoS Map Allowed by Policy : 0
```

```
PS C:\Users\LENOVO> ipconfig
```

```
Windows IP Configuration
```

```

PS C:\Users\LENOVO> ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::db30:b3c7:411a:92e1%13
    IPv4 Address. . . . . : 10.85.18.117
    Subnet Mask . . . . . : 255.255.248.0
    Default Gateway . . . . . : 10.85.16.1
PS C:\Users\LENOVO> getmac

Physical Address    Transport Name
=====
AE-39-D7-98-56-7D  \Device\NPF{B0DD9445-2EC8-4691-A2F9-F04A0B8ECAD9}
PS C:\Users\LENOVO>

```

Wi-Fi

Intel(R) Wi-Fi 6 AX203

Throughput

5 Mbps

3.6 Mbps

60 seconds

0

Send

8.0 Kbps

Receive

256 Kbps

Adapter name:

Wi-Fi

SSID:

K

Connection type:

802.11n

IPv4 address:

172.31.74.145

IPv6 address:

fe80::d305:5f2d:96c9:fd76%13

Signal strength:



⑥ Difference b/w wired & wireless speed?

Soln

Wired Network	Wireless Network
* Faster Communication	* Slower Communication
* Stable Connection	* Affected by distance / Interference
* Low latency	* Higher latency
* Dedicated Physical medium	* Shared radio medium.

→ Ethernet provides direct physical communication using network cables, resulting in higher speed & stable performance.

→ Wireless connections use radio waves which are affected by interference, distance, obstacles & multiple connected users.

Q) Role of Subnet mask in network?

→ Soln:-

A subnet mask identifies the network portion & host portion of an IP address. It helps devices determine whether another device belongs to the same network or different network.

- ① Network Segment
- ② Efficient routing
- ③ Communication b/w devices.

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OBSERVATION,

During the study of the college network infrastructure:-

- ① A centralized server room was used of network backbone
- ② Two L3 switches were connected in redundant cyclic
- ③ Managed TP-Link PoE+ switches were used in network cabinets.

Advantage of Existing Network?

- * Scalable architecture.
- * Centralized management.
- * Supports PoE+ device
- * Easy expansion of floors & classroom
- * Support both wireless & wired.
- * Redundant link for failovers.

Conclusion :-

The observed campus LAN follow a hybrid hierarchical topology consist of Tree, Extended star, Ring. The network uses managed PoE switches, wireless access point, Ethernet port & centralized server infrastructure to provide efficient wired & wireless communication across the campus.